

ABHI AREKERE

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EDUCATION

University of Texas at Austin, Cockrell School of Engineering – Austin, Texas

May 2026

Bachelor of Science in Biomedical Engineering: Cellular & Molecular Track

EXPERIENCE

Clinical Genomics Workflow Monitoring | Austin, Texas

Aug 2025 – May 2026

Industry-Sponsored Senior Design Project

- Designed structured laboratory data models (dbt: staging, intermediate, marts, UDM) to support tracking, monitoring, and analysis of clinical workflow data
- Modeled ctDNA testing workflows as a state machine system to track sample lifecycle stages, step transitions, and failure/rework events across lab processes
- Defined workflow KPIs such as dwell time, queue delays, throughput, and rework rates to identify bottlenecks and support iterative process improvements
- Built predictive models (Amazon Redshift ML) using workflow-derived features to predict turnaround time delays and sample failures
- Developed data pipelines and relational databases using PostgreSQL and Amazon Redshift with dashboard visualizations for real-time laboratory operations monitoring

Neurogenetics Lab at Texas A&M University | College Station, Texas

May 2025 – Aug 2025

Research Lab Assistant (On-site)

- Designed behavioral assays in *Drosophila* to quantify olfactory deficits associated with neurodegeneration
- Developed an agar-based assay modification to reduce locomotor bias, improving experimental control and interpretability
- Performed immunofluorescence staining and microscopy of glial cells; analyzed imaging and behavioral data using Python and R

Tumor Microenvironment at Texas A&M University | College Station, Texas

May 2024 – Aug 2024

Research Lab Intern (On-site)

- Evaluated nanomaterial-based therapies for preclinical triple-negative breast cancer using microscopy and imaging analysis
- Cultured and maintained mammalian cell lines across experimental workflows, performing passaging, media changes, and cell counting under aseptic technique in a biosafety cabinet
- Analyzed treatment response from experimental assay data using lab software, Excel, and quantitative image analysis methods

Neurogenetics Lab at Texas A&M University | College Station, Texas

May 2023 – Aug 2023

Research Lab Assistant (On-site)

- Investigated the neural regulation of sleep and feeding behavior in Mexican cavefish through ablations, dissections, and microscopy
- Executed R-based snRNA-seq analysis workflows for larval cavefish datasets using Seurat, Signac, Monocle3, and related packages for transcriptomic data analysis

Bioinformatics Lab at Texas A&M University | College Station, Texas

Jul 2020 – May 2023

Research Lab Assistant (Remote)

- Applied phylogenetic modeling and statistical analysis to evaluate evolutionary hypotheses related to chromosome fusions, fissions, range size, and genetic drift in *Carnivora*
- Developed R scripts to clean, integrate, and analyze comparative genomics datasets, including genomic, GBIF occurrence, chromosome count, and phylogenetic data; work contributed to a peer-reviewed publication in the Journal of Heredity

PUBLICATIONS

Jonika, M. M., Wilhoit, K. T., Chin, M., **Arekere, A.**, & Blackmon, H. 2024. Drift drives the evolution of chromosome number II: The impact of range size on genome evolution in *Carnivora*. *Journal of Heredity*. 115: 524-531.

TECHNICAL SKILLS

- **Languages:** Python, R, C++, SQL, MATLAB
- **Data & Analytics:** Excel, PostgreSQL, Amazon Redshift (AWS), dbt, DBeaver, workflow KPI
- **Laboratory Techniques:** Microscopy, fluorescence imaging, immunostaining, scRNA-sequencing, mammalian cell culture, GLP, SOP adherence, deviation documentation, BSL-1/BSL-2
- **CAD:** SOLIDWORKS, Fusion 360